

B.TECH
(SEM V) THEORY EXAMINATION 2022-23
OPTIMIZATION TECHNIQUES

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. What do you mean by optimization?
- b. Describe the Analytical method.
- c. Write the statement of Newton method for uni-dimensional search.
- d. Explain the region elimination methods.
- e. Explain how to resolve degeneracy in transportation problems.
- f. Discuss the basic feasible solution of LPP.
- g. Write the statement of Newton's backward formula
- h. Write the statement of Newton's divided difference formula
- i. State the Simpson's $\frac{1}{3}^{rd}$ rule.
- j. Find the initial root of the equation $x + \log x - 2 = 0$.

SECTION B

2. Attempt any three of the following: 10x3 = 30

- a. What is Fibonacci method in optimization? Explain the Fibonacci method with suitable example.
- b. Compute the real root of $x^3 - 5x + 3 = 0$ in the interval $[1, 2]$ by the Secant method by performing four iterations.
- c. Obtain the optimal of transportation cost for given data:

	D ₁	D ₂	D ₃	D ₄	supply
S ₁	19	30	50	10	7
S ₂	70	30	40	60	9
S ₃	40	8	70	20	18
demand	5	8	7	14	34

- d. Using the table given below, find the value of $f(8)$:

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

- e. Find the value of the integral $\int_0^6 \frac{1}{1+x^2} dx$ by using Simpson's $\frac{3}{8}^{th}$ rule

SECTION C

3. Attempt any one part of the following: 10x1 = 10

- a. What is multivariable optimization without constraints? How does the constrained model compare to the unconstrained optimization?
- b. What is the difference between Fibonacci and Golden section method?

4. Attempt any **one** part of the following: 10x1 = 10
- a. Explain (i) Powell's methods in multi dimensions (ii) Unconstrained one dimensional search.
- b. What is the difference between Newton and Quasi – Newton method? Is an optimizer in the family of Quasi – Newton methods?

5. Attempt any **one** part of the following: 10x1 = 10

- a. Use the graphical method to solve
- $$\text{Max } Z = 15x + 10y$$
- Subject to constraints :
- $$4x + 6y \leq 360, 3x + 0y \leq 180, 0x + 5y \leq 200 \text{ and } x, y \geq 0$$
- b. Use dual simplex method to solve: $\text{Max } Z = 3x + y$,
Subject to constraint. $x + y \geq 1, 2x + 3y \geq 2$, and $x, y \geq 0$

6. Attempt any **one** part of the following: 10x1 = 10

- a. Find the cubic polynomial which takes the following values :

x	0	1	2	3
f(x)	1	2	1	10

- b. Identify $\int_0^{\pi/2} e^{\sin x} dx$ correct to four decimal places by trapezoidal rule

7. Attempt any **one** part of the following: 10x1 = 10

- a. Using Runge–Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2, 0.4$.
- b. Find the root of $xe^x - \cos x = 0$ by Newton- Raphson method correct to the four decimal places.